

ON EXTENSIONS OF THE BINOMIAL DISTRIBUTIONS OF BERNOULLI AND POISSON

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SUMMARY

The univariate theory related to the binomial distribution of Bernoulli (i.e. the distribution of successes obtained when independent trials are made with constant probability throughout the experiment) and to the binomial distribution of Poisson (i.e. the distribution of successes when independent trials are made with probabilities varying from trial to trial) are extended to multivariate cases where events may occur zero, one or more times per trial, jointly or separately. The theory shows that in the case of fixed sampling models the variances in the Poisson cases are smaller than the variances in the Bernoullian cases. For models of the inverse Bernoullian and Poissonian type where the sampling process is stopped on obtaining a previously fixed number of failures, these results are reversed. By introducing multinomials of factorial (or hypergeometric) type similar results are shown to be true for finite populations.

1. Introduction

The introduction of methods for saving fuel and the relatively low maximum speeds imposed on the open roads of most countries have as a result that motor vehicles are moving along a given road at approximately the same speed. If we assume that in the case of an accident the probability of a fatality is dependent only on the speed, we have a system in which the probability for a fatality remains constant. If we consider the case where motor vehicles move along the roads at various speeds, we have a system where the probability of a fatality will vary from accident to accident.

In this paper we shall consider models of which the above constitutes an example. Distributions derived for a discrete system in which the probabilities for events remain constant during the process will be compared with distributions derived for a system in which the probabilities are variable. In the case of one event which can occur at most once in a single trial the relevant distributions are the binomial distributions of Bernoulli and of Poisson. In particular we shall consider models of the Bernoullian and Poissonian types where events may occur jointly or separately zero, one or more times in a single trial (e.g. allowing for the possibility of more than one fatality per accident) and then proceed to extend the ideas to truncated, negative and finite population models.

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The procedure which we shall follow is one in which the variances of the distributions in the case of a system with variable probabilities will be compared with the variances for the corresponding distributions in the case of a system with constant probabilities and where the relevant probabilities for the two systems are chosen in such a way that the mean values of the distributions which are compared, are the same.

Starting with the probability for x successes in n trials where p is the probability for a success in a single trial we know that the probability generating function (p.g.f.) is $(q+pt)^n$, where $q=1-p$, and further that,

$$(1.1) \mu'_1 = np, \sigma^2(B) = npq,$$

where the symbol B indicates that we are dealing with the Bernoullian case or its extensions. The binomial distribution of Poisson is obtained when the probability differs from trial to trial. If p_i is the probability of a success in the i -th trial then the probability of x successes after n trials is given by the coefficient of t^x in $\prod_{i=1}^n (q_i + p_i t)$ with

$$(1.2) \quad \begin{aligned} \mu'_1 &\doteq np; \sigma^2(P) = \sigma^2(B) - \sum_{i=1}^n (p_i - p)^2 \\ &= \sigma^2(B) - \sum_{j < u} (p_j - p_u)^2/n, \end{aligned}$$

where $p = \sum_{i=1}^n p_i/n$ is the "mean" probability and $\sigma^2(B)$ is defined as in (1.1). We shall use the symbol P to indicate reference to the binomial of Poisson and its extensions. The variance in the case of the binomial distribution of Poisson is thus smaller than the variance of the binomial distribution of Bernoulli with the same mean value.

The expressions (1.2) resemble the formulae for stratification with respect to the variable but cannot be written down from standard theory since stratification in the case under consideration is in respect of the probabilities and not in respect of the values of the variable.

2. Generalized multivariate distributions of Bernoullian and Poissonian type

2.1 The Bernoullian case

We shall now fix our attention on the case of k events $E_1, \dots, E_k$ which can occur zero, one or more times separately or jointly in each trial. Let $p_{i_1, \dots, i_k}$ be the constant probability that in a single trial the events E_j will occur i_j times ($i_j = 0, \dots, h_j$), $j=1, \dots, k$. In this case the probability that in n trials, E_j , ($j=1, \dots, k$), will occur x_j times is given

by the coefficient of $t_1^{x_1} t_2^{x_2} \dots t_k^{x_k}$ in the expansion of the multinomial expression

$$(2.1) \left(\sum_{i_1=0}^{h_1} \dots \sum_{i_k=0}^{h_k} p_{i_1, \dots, i_k} t_1^{i_1} \dots t_k^{i_k} \right)^n.$$

From (2.1) follows immediately, by putting $t_j=1$ for all $j \neq s$, that the marginal distribution of x_s is a univariate multinomial distribution with probability generating function

$$(2.2) \left(\sum_{i_s=0}^{h_s} p_{i_s} t_s^{i_s} \right)^n$$

where p_{i_s} is the sum of all the $p_{i_1, \dots, i_k}$ for which i_s is the s -th suffix.

From (2.2) follows, after substituting $t_s=1+\alpha_s$ and by using the well-known relation between moments and factorial moments, that

$$(2.3) \mu_s \equiv E(x_s) = n \sum_{i_s=0}^{h_s} i_s p_{i_s} \text{ and } \sigma_s^2(B) = n \left(\sum_{i_s=0}^{h_s} i_s^2 p_{i_s} - \left(\sum_{i_s=0}^{h_s} i_s p_{i_s} \right)^2 \right).$$

Similarly we have that the marginal distribution of x_s and x_r is generated by

$$(2.4) \left(\sum_{i_s=0}^{h_s} \sum_{i_r=0}^{h_r} p_{i_s, i_r} t_s^{i_s} t_r^{i_r} \right)^n$$

where p_{i_s, i_r} is the sum of all the $p_{i_1, \dots, i_k}$ which have an i_s and an i_r as the s -th and r -th suffix. From (2.4) we have amongst others that the central product moment of x_s and x_r is

$$(2.5) \rho_{sr} \sigma_s \sigma_r = E(x_s - \mu_s)(x_r - \mu_r) = n \sum_{i_s=0}^{h_s} \sum_{i_r=0}^{h_r} i_s i_r (p_{i_s, i_r} - p_{i_s} p_{i_r}).$$

The central product moments are written down although they are not used in comparing the relevant distributions. They are, however, necessary when multivariate normal approximations of the discrete distributions dealt with in this paper are considered.

2.2 The Poissonian case

We now consider the case where in the model (under 2.1) $p_{1; i_1, \dots, i_k}$ is the constant probability during the first n_1 trials, while $p_{2; i_1, \dots, i_k}$ is the constant probability during the next n_2 trials and so on until $p_{\ell; i_1, \dots, i_k}$ which is the constant probability during the last n_ℓ trials so that $n = \sum_{j=1}^{\ell} n_j$, and in $p_{j; i_1, \dots, i_k}$ the suffix $i_r = 0, \dots, h_{i_r}$.

The probability generating function is now given by

$$(2.6) \prod_{j=1}^{\ell} \left(\sum_{i_1=0}^{h_{i_1}} \dots \sum_{i_k=0}^{h_{i_k}} p_{j; i_1, \dots, i_k} t_1^{i_1} \dots t_k^{i_k} \right)^{n_j}.$$

Substituting $t_j = \exp(\alpha_j)$ and taking the logarithm to find the cumulant generating function we see that the contribution of each sub-group of n_j trials may be added to determine the cumulants. From (2.3) and (2.5) we may now write down:

$$(2.7) \quad \mu_s \equiv E(x_s) = \sum_{j=1}^{\ell} n_j \sum_{i_s=0}^{h_{js}} i_s p_{j,i_s}$$

$$(2.8) \quad \sigma_s^2(P) = \sum_{j=1}^{\ell} n_j \left(\sum_{i_s=0}^{h_{js}} i_s^2 p_{j,i_s} - \left(\sum_{i_s=0}^{h_{js}} i_s p_{j,i_s} \right)^2 \right) \quad \text{and}$$

$$(2.9) \quad \rho_{sr} \sigma_s \sigma_r = \sum_{j=1}^{\ell} n_j \sum_{i_s=0}^{h_{js}} \sum_{i_r=0}^{h_{jr}} i_s i_r (p_{j,i_s,i_r} - p_{j,i_s} p_{j,i_r})$$

where p_{j,i_s} and p_{j,i_s,i_r} are defined as in Section 2.1 but with respect to $p_{j,i_1,\dots,i_k}$. Similar to the univariate case dealt with in Section 1 we shall now compare $\sigma_s^2(B)$ and $\sigma_s^2(P)$ by using (2.3) and (2.8). In order to do the comparison, let us define

$n p_{i_1,\dots,i_k} \equiv \sum_{j=1}^{\ell} n_j p_{j,i_1,\dots,i_k}$, which gives on summation

$$(2.10) \quad n p_{i_s} = \sum_{j=1}^{\ell} n_j p_{j,i_s}$$

so that (2.7) may be written as $\mu_s = n \sum_{i_s=0}^{h_s} i_s p_{i_s}$ where $h_s \equiv \max h_{js}$.

This expression shows that the mean value in the Poisson case is equal to the mean value in the Bernoullian case provided that p_{i_s} is defined as in (2.10).

From (2.3) and (2.8) leaving out the suffix s and writing i , h_j and h for i_s , h_{js} and h_s we have:

$$\begin{aligned} \sigma^2(P) &= \sigma^2(B) + \sum_{j=1}^{\ell} n_j \left\{ \sum_{i=0}^{h_j} i^2 p_{j,i} - \left(\sum_{i=0}^{h_j} i p_{j,i} \right)^2 \right\} - n \left\{ \sum_{i=0}^h i^2 p_i - \left(\sum_{i=0}^h i p_i \right)^2 \right\} \\ &= \sigma^2(B) + \sum_{i=0}^h i^2 \left(\sum_{j=1}^{\ell} n_j p_{j,i} - n p_i \right) - \left(\sum_{j=1}^{\ell} n_j \left(\sum_{i=0}^{h_j} i p_{j,i} \right)^2 - (1/n) \left(\sum_{i=0}^h i n p_i \right)^2 \right). \end{aligned}$$

The second term on the right-hand side is zero by (2.10). Again, using (2.10) the last term is equal to

$$\begin{aligned} (1/n) \left[\sum_{j=1}^{\ell} n_j (n_1 + \dots + n_{\ell}) Z_j^2 - \left(\sum_{j=1}^{\ell} n_j Z_j \right)^2 \right] \text{ where } Z = \sum_{i=0}^{h_j} i p_{j,i} \\ = (1/n) \sum_{j < u} n_j n_u (Z_j - Z_u)^2. \end{aligned}$$

Introducing the suffix s again we have:

$$(2.11) \sigma_s^2(P) = \sigma_s^2(B) - (1/n) \sum_{j < u} n_j n_u \left(\sum_{i_s=0}^{h_{js}} i_s p_{j i_s} - \sum_{i_s=0}^{h_{us}} i_s p_{u i_s} \right)^2.$$

The result that $\sigma_s^2(B)$ is larger than $\sigma_s^2(P)$ may be compared with the well-known result that for samples of fixed size the variance for random samples is generally larger than for stratified random samples. In the process leading up to (2.11) the stratification is with respect to probabilities. The exact form (2.11) and its interpretation is interesting.

It is clear from (2.3) and (2.7) that $\sum_{i_s=0}^{h_{js}} i_s p_{j i_s}$ is the expected number of successes of event E_s per trial. The interpretation of the difference $\sigma_s^2(B) - \sigma_s^2(P)$ is thus straight-forward and can be estimated by the sum of squares shown in (2.11) provided that the mean number of successes in each trial can be estimated for each sub-group.

3. Truncation at zero

If we ignore the zero value in the binomial distribution of Bernoulli we obtain the truncated Bernoullian distribution with point of truncation at zero. When it is required to refer to this distribution or its extensions we shall use the symbol TB. The probability function is given by $P(X=x) = \binom{n}{x} p^x q^{n-x} / (1-q^n)$ where $q = 1-p$, and the probability generating function (p.g.f.) by $f(t) = ((q+pt)^n - q^n) / (1-q^n)$.

We now proceed to study the truncated distribution (with truncation of zero) for the multivariate model which has lead to (2.1) above. In this case the p.g.f. is given by

$$(3.1) ((A + \sum_{i_1 + \dots + i_k > 0} p_{i_1, \dots, i_k} t_1^{i_1} \dots t_k^{i_k})^n - A^n) / (1-A^n) \text{ where } A = p_{0, \dots, 0}.$$

After algebraic simplification the moments of the first and second order follows from (3.1) as

$$(3.2) E(x_s) = n \sum_{i_s=0}^{h_s} i_s p_{i_s} / (1-A^n)$$

$$(3.3) \sigma_s^2(TB) = n \left\{ \sum_{i_s=0}^{h_s} i_s^2 p_{i_s} - \left\{ (1 + n \overline{1A^n}) / (1-A^n) \right\} \left(\sum_{i_s=0}^{h_s} i_s p_{i_s} \right)^2 \right\} / (1-A^n)$$

and

$$(3.4) \rho_{rs} \sigma_r \sigma_s = n \left\{ \sum_i \sum_r i_s i_r p_{i_s, i_r} - ((1 + n \overline{1A^n}) / (1-A^n)) \sum_{i_s} \sum_{i_r} i_s i_r p_{i_s} p_{i_r} \right\} / (1-A^n).$$

We proceed now to the truncated case corresponding to the model of variable probabilities considered in Section 2.2 which has lead to the p.g.f. (2.6). For the

truncated Poissonian distribution (abbreviated by TP) the p.g.f. is obtained as the product of ℓ factors of the form (3.1) where in the j -th factor ($j=1, \dots, \ell$), A is replaced by $A_j = p_{j0}, \dots, p_{jn_j}$ and n is replaced by n_j . The first and second order moments $E(x_s)$ and $\sigma_s^2(\text{TP})$ then follow from (3.2) and (3.3) by summing w.r.t. the suffix j .

Let us now define probabilities $p_{i_1, \dots, i_k}$ by

$$(3.5) \quad n p_{i_1, \dots, i_k} / (1-A^n) = \sum_{j=1}^{\ell} n_j p_{ji_1, \dots, i_k} / (1-A_j^{n_j}).$$

Similar to the procedure in Section 2 we compare the variances $\sigma_s^2(\text{TB})$ and $\sigma_s^2(\text{TP})$. Writing $p'_{js} = p_{js} / (1-A_j^{n_j})$ and using the same method as in Section 2, we find after algebraic simplification that

$$(3.6) \quad \sigma_s^2(\text{TP}) = \sigma_s^2(\text{TB}) - \sum_{j < u} (n_j n_u / n) \left(\sum_{i_s=0}^{h_{js}} i_s p'_{js} - \sum_{i_s=0}^{h_{js}} i_s p'_{us} \right)^2 + O(A_j)^{n_j}.$$

The interpretation of (3.6) is in general the same as that of (2.11) above but for small values of the n_j a closer investigation may be necessary.

The results of Sections 2 and 3 refer to models with fixed sample size n . In both instances the variances in the Bernoullian case were larger than in the Poissonian case. We shall now proceed to study a particular model where the sample sizes are not previously fixed.

4. The case of negative multinomials

We shall use as far as possible the same notation as in the previous section. In this section we shall compare the model of constant probabilities with the model of variable probabilities for a sampling process which is terminated on obtaining a previously fixed number of failures (that is, none of the events $E_1, \dots, E_k$ have occurred a fixed number of times).

Using the notation of Section 2.1 the p.g.f. for the case where the trials are stopped on obtaining the m -th failure is given by

$$(4.1) \quad A^m \left(1 - \sum_{i_1=0}^{h_1} \dots \sum_{i_k=0}^{h_k} p_{i_1, \dots, i_k} t_1^{i_1} \dots t_k^{i_k} \right)^{-m} \quad \text{where } \sum_{r=1}^k i_r \neq 0.$$

This expression follows directly from first principles (see Wiid (1958)). We shall refer to this model as the negative Bernoullian (NB) case.

Next, using the notation and model of Section 2.2 the probability generating function obtained when the trials in the j -th sub-group are stopped on obtaining the m_j -th failure, is given by

$$(4.2) \quad \prod_{j=1}^{\ell} A_j^{m_j} \left(1 - \sum_{i_1=0}^{h_{j1}} \dots \sum_{i_k=0}^{h_{jk}} p_{ji_1, \dots, i_k} t_1^{i_1} \dots t_k^{i_k} \right)^{-m_j} \quad \text{where } \sum_{r=1}^k i_r \neq 0.$$

We shall refer to this model as the negative Poissonian (NP) case.

From (4.1) we find after substituting $t_i = \exp(\alpha_i)$, $i=1, \dots, k$ and expanding this cumulant generating function as a power series in α_i , ($i=1, \dots, k$) that

$$(4.3) \quad E(x_s) = m \sum_{i_s=0}^{h_s} i_s p_{i_s} / A$$

$$(4.4) \quad \sigma_s^2(\text{NB}) = m \left(\left(\sum_{i_s=0}^{h_s} i_s p_{i_s} \right)^2 + A \sum_{i_s=0}^{h_s} i_s^2 p_{i_s} \right) / A^2$$

and

$$(4.5) \quad \rho_{sr} \sigma_s \sigma_r = m \left(\sum_{i_s=0}^{h_s} \sum_{i_r=0}^{h_r} i_s i_r (p_{i_s} p_{i_r} + A p_{i_s} p_{i_r}) \right) / A^2.$$

The expected sample size is given by

$$(4.6) \quad m + \left(\sum_{i_1 + \dots + i_k > 0} m p_{i_1, \dots, i_k} \right) / A = m/A$$

Starting with (4.2) and proceeding exactly as in Section 2 the moments $E(x_s)$ and $\sigma_s^2(\text{NP})$ can be written down from (4.3) and (4.4) by summing over the suffix j .

The expected sample size within sub-group "j" is given by m_j/A_j . If we now define probabilities $p_{j i_1, \dots, i_k}$ in terms of the $p_{j i_1, \dots, i_k}$ by

$$(4.7) \quad m p_{i_1, \dots, i_k} / A \equiv \sum_{j=1}^{\ell} m_j p_{j i_1, \dots, i_k} / A_j$$

it follows by putting $i_1 = \dots = i_k = 0$ that $m = \sum_{j=1}^{\ell} m_j$, and after summation w.r.t. all the values of $i_1, \dots, i_k$ that, $m/A = \sum_{j=1}^{\ell} m_j / A_j$. Each $p_{i_1, \dots, i_k}$ in (4.7) is therefore a weighted mean obtained from the $p_{j i_1, \dots, i_k}$ ($j=1, \dots, \ell$) by using the expected sample sizes of each sub-group as weights.

By using (4.7) we see that the expected values in both models become the same. We note that (4.3), (4.4) and (4.5) can be obtained from (2.3) and (2.5) by substituting $-m$ for n and $-p_{i_1, \dots, i_k} / A$ for $p_{i_1, \dots, i_k}$. Using (2.11) we may thus write down immediately that

$$(4.8) \quad \sigma_s^2(\text{NP}) = \sigma_s^2(\text{NB}) + \sum_{j < u} (m_j m_u / m) \left(\sum_{i_s=0}^{h_{js}} i_s p_{j i_s} / A_j - \sum_{i_s=0}^{h_{us}} i_s p_{u i_s} / A_u \right)^2$$

The result (4.8) shows that contrary to the case of fixed sample size, the variance in the negative Poissonian case is larger than the variance of the corresponding negative Bernoullian case.

5. The case of finite populations

Let us start again with an example and consider N motor vehicles moving along

$(k+1)$ different roads such that there are Np_i , $i=0, \dots, k$, along road "i". The vehicles keep moving until there has been a total of n accidents each accident involving only one vehicle. The probability that x_i accidents have occurred on road "i" out of the total number of n accidents, is then given by the well-known multivariate hypergeometric distribution

$$(5.1) \quad P(X=x) = (n! / \prod_{i=0}^k x_i!) \left(\prod_{i=0}^k (Np_i)^{x_i} \right) / N^{(n)},$$

where X is a random vector, $x=(x_1, \dots, x_k)$ its numerical value, $x_0 \equiv n - \sum_{i=1}^k x_i$, $\sum_{i=0}^k p_i = 1$ and $a^{(r)} \equiv a(a-1) \dots (a-r+1)$. The form (5.1) of the hypergeometric distribution is used on purpose. When $N \rightarrow \infty$ we see immediately that the r.h.s. of (5.1) becomes

$$(5.2) \quad (n! / \prod_{i=0}^k x_i!) \prod_{i=0}^k p_i^{x_i}$$

which is the well-known multinomial distribution and is the general term in the expansion of the multinomial $(p_0 + \dots + p_k)^n$ as a power series. Similarly (5.1) is the general term in the expansion of the factorial multinomial

$$(Np_0 + \dots + Np_k)^{(n)} / N^{(n)}$$

into a factorial power series by the well-known method of Van der Monde. If we introduce the convention that $(Np_i t_i)^{(r)}$ will mean $(Np_i)^{(r)} t_i^r$ then the p.g.f. of (5.1) can be written as

$$(5.3) \quad (Np_0 + Np_1 t_1 + \dots + Np_k t_k)^{(n)} / N^{(n)},$$

and can be compared with the special case of (2.1) when the k events are mutually exclusive and can occur only once in each trial, so that (2.1) has the form

$$(5.4) \quad (p_0 + p_1 t_1 + \dots + p_k t_k)^n.$$

Comparing (5.3) with (5.4) we clearly see that the finite population cases corresponding to the models considered in Sections 2, 3 and 4 will differ only in that the multinomials entering into the expressions will have to be replaced by factorial multinomials using the above convention. Although (5.4) and (5.3) only deal with very special cases it is easily verified that the statement is generally true. Only the special case will be considered further.

In the above example a single sample of n items is taken without replacement from a finite population of size N subdivided into classes $(Np_0, \dots, Np_k)$ where $\sum_{i=0}^k p_i = 1$.

Let us name this model the factorial Bernoullian (FB) model and proceed to compare it with a model where the sample size n is taken in stages. In stage one a sample of n_1 items is taken without replacement from the finite population N_1 subdivided as

$(N_1 p_{1,0}, \dots, N_1 p_{1,k})$; in stage two a sample of n_2 items is taken without replacement from the remaining $N_2 = N_1 - n_1$ items now subdivided as $(N_2 p_{2,0}, \dots, N_2 p_{2,k})$ and so on until at stage ℓ a sample of n_ℓ items is taken without replacement from the remaining $N_\ell = N_1 - \sum_{j=1}^{\ell-1} n_j$ subdivided as $(N_\ell p_{\ell,0}, \dots, N_\ell p_{\ell,k})$ where $n = \sum_{j=1}^{\ell} n_j$. (Compare the case where in the previously mentioned example the proportion of motor vehicles on the $(k+1)$ roads change from day to day.) We shall refer to this model as the factorial Poissonian (FP) case. Since the sampling at the various stages is done independently the p.g.f. for obtaining x_i successes from class "i", $i=0, \dots, k$ is given (using the abovementioned convention), by

$$(5.6) \quad \prod_{j=1}^{\ell} (N_j p_{j,0} + N_j p_{j,1} t_1 + \dots + N_j p_{j,k} t_k)^{(n_j)} / N^{(n_j)}$$

For the distribution (5.1) the following results are known (see Steyn (1955) or Johnson and Kotz (1969), p. 144):

$$(5.7) \quad E(x_s) = np_s$$

$$(5.8) \quad \sigma_s^2(\text{FB}) = np_s(1-p_s)(N-n) / (N-1)$$

$$(5.9) \quad \rho_{sr} \sigma_s \sigma_r = -np_s p_r (N-n) / (N-1)$$

For the distribution defined by (5.6) the corresponding moments are obtained from (5.7), (5.8) and (5.9) by summation over the suffix j after substituting N_j , n_j and $p_{j,s}$ for N , n and p_s .

Similar to the procedure used in the previous sections we now define p_s by $np_s \equiv \sum_{j=1}^{\ell} n_j p_{j,s}$ so that $n = \sum_{j=1}^{\ell} n_j$ and the expected values in both models are the same. Writing N for N_1 and using $(N_j - n_j)/(N_j - 1) = 1 + O(n_j/N)$ we have by (2.11) that

$$(5.10) \quad \sigma_s^2(\text{FP}) = \sigma_s^2(\text{FB}) - (1/n) \sum_{j < u} n_j n_u (p_j - p_u)^2 + O(n/N)$$

In general it follows that the variance in the factorial Poissonian case will be smaller than the variance in the factorial Bernoullian case if n is small compared to N . Further extensions are now a matter of straightforward manipulation.

6. Limiting forms

If we start with the probability functions of the previous sections and then proceed, by substitution and taking logarithms, to the cumulant generating functions, it becomes clear that the cumulants for all the distributions will be of $O(n)$ so that in standard units all cumulants except those of the second order will vanish when $n \rightarrow \infty$. All the distributions dealt with in this paper therefore tend to the multivariate normal distribution.

REFERENCES

- JOHNSON, N.L. and KOTZ, S. (1969). *Discrete Distributions*. Houghton Mifflin Company, Boston, Mass.
- STEYN, H.S. (1955). On discrete multivariate probability functions of hypergeometric type. *Nederl. Akad. Wetensch. Proc. Sec. A* 58(5) 588-595.
- STEYN, H.S. (1963). On approximations for the distributions obtained from multiple events. *Nederl. Akad. Wetensch. Proc. Sec. A* 66(1) 85-96.
- WIID, A.J.B. (1958). On the moments and regression equations of the fourfold negative factorial binomial distributions. *Proc. Roy. Soc. Edinburgh Sec. A* LXV(1) 29-34.

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